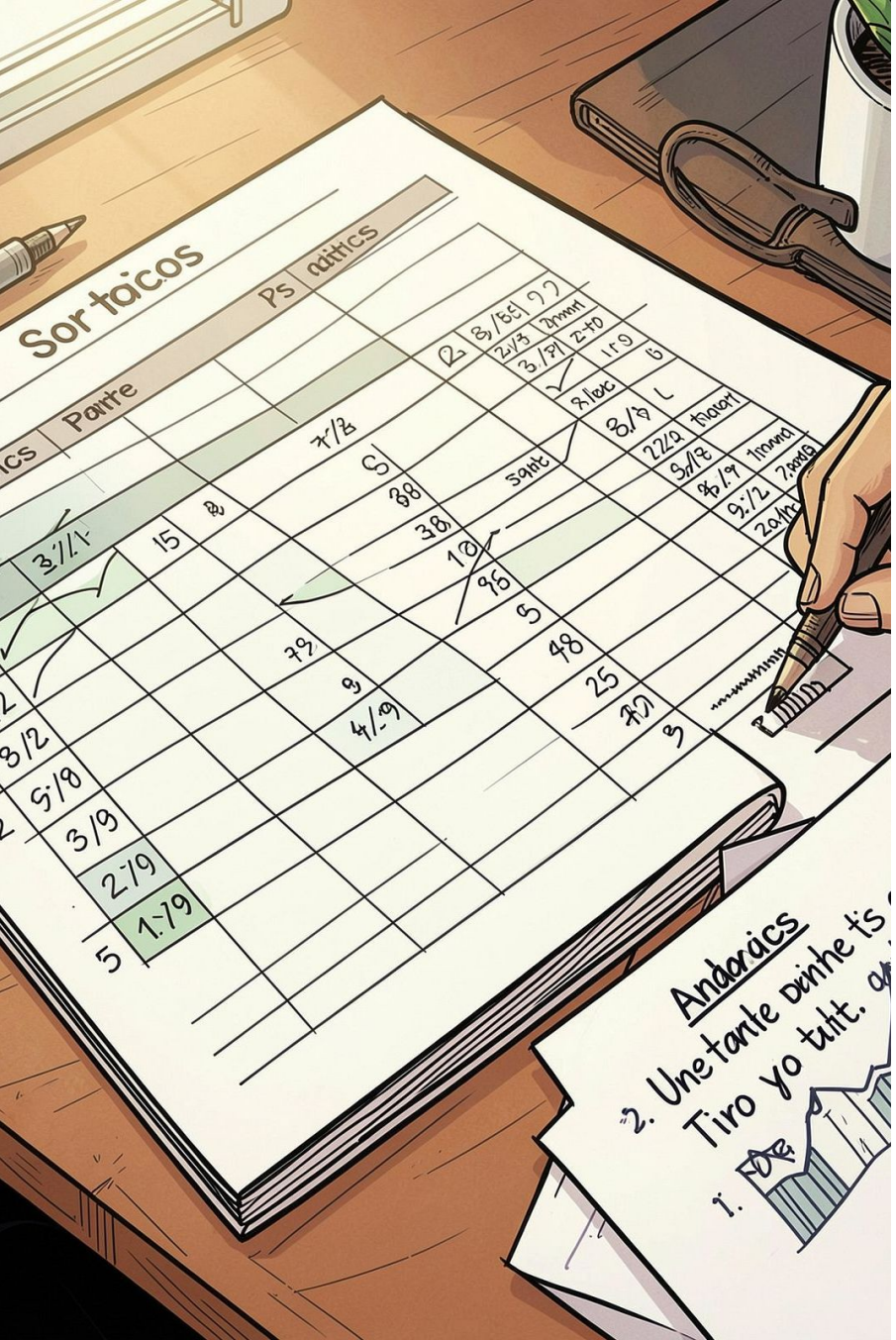


Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: insights to guide strategy

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Dataset Snapshot

Scale

Rows: 3,900 • Columns: 18

Key Features

Demographics, Purchase details, Behavior flags

Missing Data

37 values in Review Rating

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Data Preparation (Python)

Load → Inspect → Clean

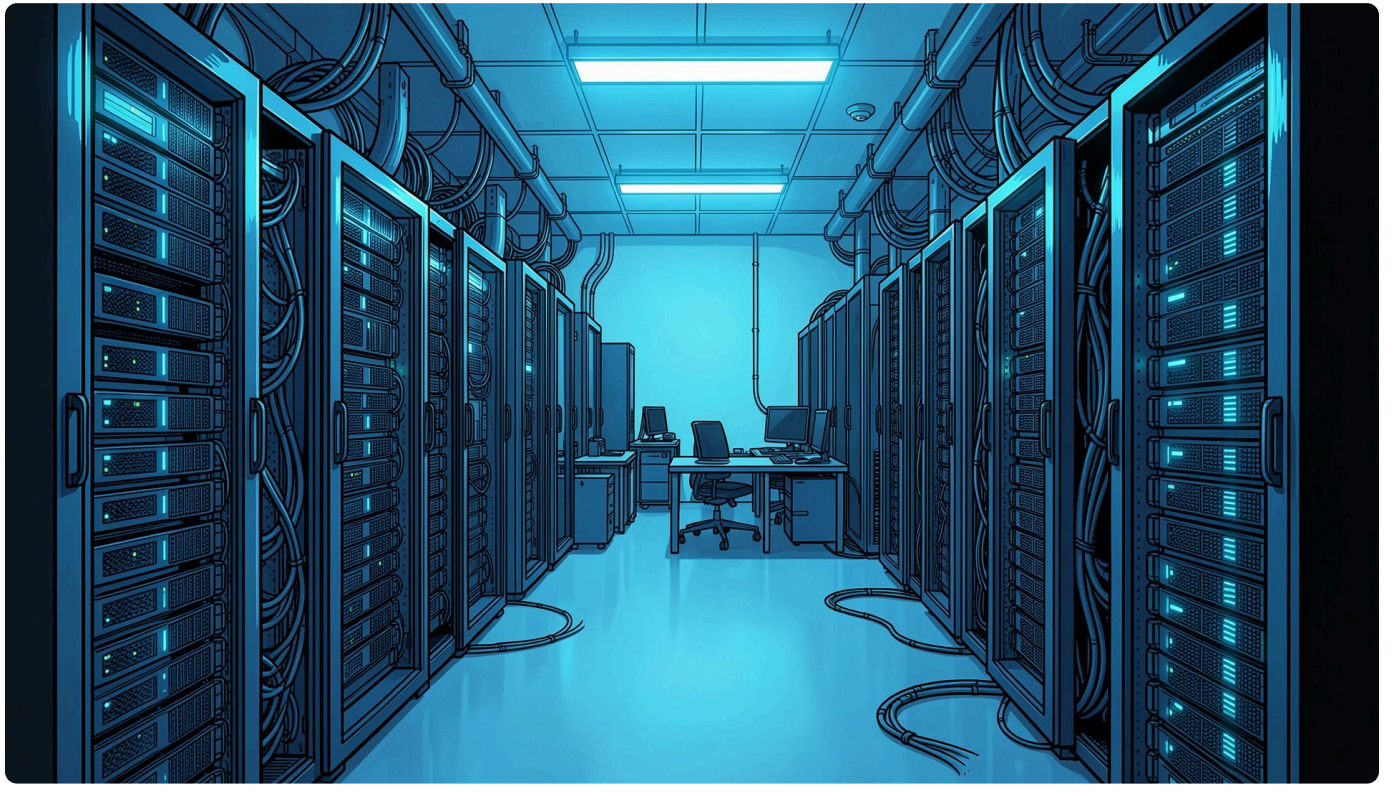
- df.info(), describe()
- Impute review_rating by category median
- Rename columns to snake_case

Feature engineering

- age_group (bins)
- purchase_frequency_days
- Dropped promo_code_used (redundant)



Database Integration



mysql

Cleaned DataFrame loaded to SQL for structured analysis

Why

Scalable queries ▪ joins ▪ production-ready reporting

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Revenue Highlights (SQL)

By Gender

Female: 75,191 ▪ Male: 157,890

Shipping Type

Standard avg: 58.46 ▪ Express avg: 60.48

Subscribers

Yes: 1,053 (avg 59.49, rev 62,645) ▪ No: 2,847 (avg 59.87, rev 170,436)

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Top Products & Ratings



Top 5 by avg rating

- Gloves — 3.86
- Sandals — 3.84
- Boots — 3.82
- Hat — 3.80
- Skirt — 3.78

Top items by orders (examples): Blouse, Pants, Jewelry, Sunglasses, Sandals

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Discounts & High-Spenders



High-spending users who used discounts — 839 rows identified

Top discount-dependent items (% discounted):

- Hat 50.00%
- Sneakers 49.66%
- Coat 49.07%
- Sweater 48.17%
- Pants 47.37%

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Customer Segments



Loyal

3,116 customers



New

83 customers



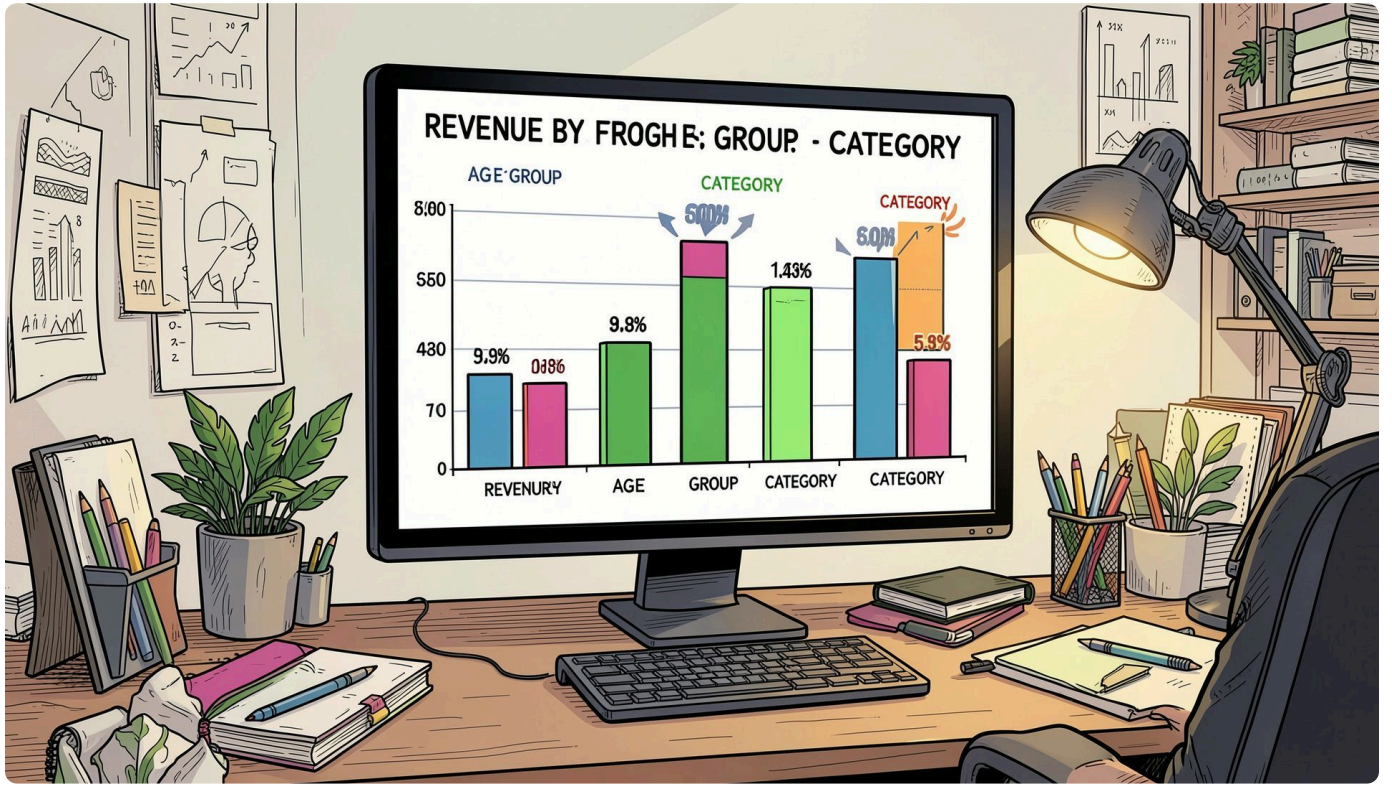
Returning

701 customers

Repeat buyers (>5 purchases): Subscribers vs non — No: 2,518 ▪ Yes: 958

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Revenue by Age Group & Category



Revenue by age group

- Young Adult — 62,143
- Middle-aged — 59,197
- Adult — 55,978
- Senior — 55,763

Revenue by category (approx)

- Clothing ~100K
- Accessories ~75K
- Footwear ~35K
- Outerwear ~20K

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to move them to Loyal

Review Discount Policy

Balance short-term sales with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Product Positioning

Feature top-rated & best-selling items in campaigns

Dashboard: Power BI interactive report created



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